RentaMovil: Vehicle Rental Platform

TECHNICAL MANUAL

Version 2.0

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Version control

Version	Date	Change description	Responsible
1.0	09/07/2026	Initial creation of the technical manual.	Development team
2.0	06/10/2026	Update to match the current architecture: microservices (API Gateway, iam, fleet-maintenance, booking-reservation), SQL Server database split by schema with Liquibase, data dictionary, endpoints, security, version control, maintenance and troubleshooting.	Development team

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1. Introduction

This manual describes how RentaMovil, the vehicle rental platform, is built: its architecture, the technologies it uses, the code organization, the database, the technical modules, the application programming interface (API), security, and the recommendations for maintaining and extending it.

It is aimed at developers, technical support staff and instructors who need to understand, maintain or continue the system. It assumes knowledge of Java, Spring Boot, React, SQL Server, Git, Docker and HTTP. It does not explain how the system is used from its screens (that is in the User Manual and the Administrator Manual) nor the installation step by step (Installation Manual).

1.1 Conventions

- Commands, paths, file names, variables and classes are written in monospace font inside grey blocks or inline in the text (for example, application.properties, APP_IAM_JWT_EXPIRY, /auth/login).
- API paths are written as the client calls them through the gateway (port 8080), unless otherwise stated.
- The code and technical documentation in the repositories are in English; what the user sees (messages, screens) is in Spanish. This is a decision recorded in ADR-001, not an oversight.
- Everything described was verified against the repository code on 6 October 2026 (branches listed in section 4.2).

2. Objective

Document RentaMovil's architecture, technologies, project structure, database, modules, API, dependencies, security and maintenance recommendations, so that another development team can understand the system, support it and continue building it.

3. Technical scope

The manual covers:

- The web frontend (customer and administration) and the mobile app.
- The API Gateway and the implemented microservices: rtm-iam, rtm-fleet-maintenance and rtm-booking-reservation.
- The SQL Server database, its schemas, tables, relationships and Liquibase migrations.
- The mock server that temporarily replaces the payment and GPS services.
- Security, version control, maintenance, troubleshooting and reference commands.

It does not cover the detail of each request and response body (that is in the OpenAPI contracts in rtm-docs/07-api/contracts), the planned services that do not exist yet (payment-billing and telemetry-gps) beyond their place in the architecture, nor the full installation procedure.

4. General system description

RentaMovil lets customers search for available vehicles by branch and dates, reserve them with optional insurance, pay by bank transfer with a receipt, and check their reservations and notifications. Administrators manage the fleet, maintenance, branches, insurance, bank accounts, users, payment validation, vehicle pickup and return, and GPS tracking.

It is a client-server application: two clients (web and mobile) consume, through an API Gateway, a Java microservices backend that share a single SQL Server database, split by schema.

4.1 Repositories

Table 1

System repositories

Repository	Location	Contents
RentaMovil	github.com/RentaMovil/RentaMovil (mirror: nicolerv18/RentaMovil)	Web frontend (web/Front-end), mobile app (app_mobile/Frontend), mock server and functional documentation.
rtm-api-gateway	github.com/RentaMovil/rtm-api-gateway	Single entry point: routing, JWT verification, CORS, rate limiting and traceability.
rtm-iam	github.com/RentaMovil/rtm-iam	Identity and access: registration, login, JWT, sessions, profile, users and roles.
rtm-iam-db	github.com/RentaMovil/rtm-iam-db	Liquibase migrations for the iam schema and the shared.audit table; SQL Server container.
rtm-fleet-maintenance	github.com/RentaMovil/rtm-fleet-maintenance	Fleet: vehicles, branches, maintenance and reference catalogs.

Repository	Location	Contents
rtm-fleet-maintenance-db	github.com/RentaMovil/rtm-fleet-maintenance-db	Liquibase migrations for the fleet schema.
rtm-booking-reservation	github.com/RentaMovil/rtm-booking-reservation	Reservations, rental execution (pickup and return), insurance plans and notifications.
rtm-booking-db	github.com/RentaMovil/rtm-booking-db	Liquibase migrations for the booking schema.
rtm-docs	github.com/code-sena/rtm-docs	Governance, domain, architecture (ADR), data, OpenAPI contract and operations documentation.

4.2 Implementation status

Verified branches: RentaMovil develop; rtm-api-gateway feat/booking-routes; rtm-iam feat/ConectBooking; rtm-booking-reservation develop; rtm-fleet-maintenance develop; *-db repositories develop; rtm-docs main.

Table 2

Status of each component

Component	Status	Note
Web frontend	Operational	Connected to the gateway for auth, users, vehicles, branches, maintenance, reservations, rentals, insurance and notifications. Payments, bank accounts and GPS are served from the mock.
Mobile app	Partial	Authentication and notifications point to the old mock (port 3001, /auth/* routes); branches, insurance, reservations and payments use local data. It is not connected to the gateway yet.
rtm-api-gateway	Operational	Routes the five services; the two planned ones respond 503 until they exist.

Component	Status	Note
rtm-iam	Operational	All identity, profile and user administration operations.
rtm-fleet-maintenance	Operational (develop branch)	Vehicles, inventory, availability, branches, maintenance and catalogs.
rtm-booking-reservation	Operational with pending dependencies	Creating a reservation responds 409 FLEET_SERVICE_UNAVAILABLE because its fleet adapter is still provisional. Confirming a reservation requires payment-billing or a manual event at /internal/events.
payment-billing (:3004)	Planned	Replaced by the mock (/payments, /bank-accounts).
telemetry-gps (:3005)	Planned	Replaced by the mock (/gps, /locations).

5. Technologies used

Table 3

Technologies and versions

Component	Technology	Version / detail
Web frontend	React (Meta Platforms, Inc., n.d.) + Vite (Vite, n.d.)	React 19.2, Vite 8, React Router 7, react-hook-form 7, i18next 26, Leaflet 1.9 / react-leaflet 5, Recharts 3, GSAP 3, react-dropzone 15
Mobile app	Expo (Expo, n.d.) + React Native (TypeScript)	Expo SDK 54, React Native 0.81.5, React 19.1, expo-router 6, Zustand 5, expo-secure-store 15, TypeScript 5.9
Backend	Java + Spring Boot (VMware, Inc., n.d.-a)	Java 21; Spring Boot 4.1.1 (iam, fleet, booking); Spring Boot 4.0.8 (gateway)
API Gateway	Spring Cloud Gateway (VMware, Inc., n.d.-b) (WebMVC)	Spring Cloud 2025.1.3
Persistence	Spring Data JPA (VMware, Inc., n.d.-c) + Hibernate	ddl-auto=validate; JDBC time zone UTC; mssql-jdbc driver
Security	Nimbus JOSE + JWT (Connect2id, n.d.; Jones et al., 2015); Spring Security Crypto	nimbus-jose-jwt 10.10 (RS256); BCrypt (12 rounds)
Database	Microsoft SQL Server (Microsoft, n.d.)	SQL Server 2022 (image mcr.microsoft.com/mssql/server:2022-latest, Developer edition)
Migrations	Liquibase (Liquibase, n.d.) (Maven plugin, Apache Software Foundation, n.d.)	Liquibase 5.0.4
Build	Maven (Apache Software Foundation, n.d.) / npm (OpenJS Foundation, n.d.)	Maven 3.9; Node.js 22 (image node:22-alpine)

Component	Technology	Version / detail
Containers	Docker (Docker Inc., n.d.) and Docker Compose	Shared external network rentamovil; images eclipse-temurin:21-jre
API mock	json-server + Express middleware	json-server 0.17, jsonwebtoken 9, bcryptjs 3 (mock-server.cjs, port 3100)
Image storage	Cloudinary	Direct upload from the browser (vehicle photos, profile and receipts)
API contracts	OpenAPI 3 (OpenAPI Initiative, 2021)	rtm-docs/07-api/contracts/openapi (one file per service + _shared.yaml)
Version control	Git (Git, n.d.) + GitHub	RentaMovil organization; branches develop → qa → staging → main; Conventional Commits (Conventional Commits, n.d.)
Diagrams	Mermaid, PlantUML, Graphviz	rtm-docs and Documentacion/Diagramas UML

5.1 Dependencies per component

Table 4

Dependency management per component

Component	Dependency file	Manager
rtm-api-gateway, rtm-iam, rtm-fleet-maintenance, rtm-booking-reservation	pom.xml	Maven 3.9
rtm-iam-db, rtm-fleet-maintenance-db, rtm-booking-reservation-db (*-db repositories)	pom.xml (Liquibase plugin)	Maven 3.9 + Liquibase 5.0.4
web/Front-end	package.json	npm (Node.js 22)
app_mobile/ Frontend	package.json	npm (Expo SDK 54)

Each repository declares its own dependencies in its pom.xml or package.json file; since this is a microservices architecture with independent repositories, there is no single dependency file for the whole system. The main dependencies per component are listed in Table 3 (section 5):

- Backend (Java + Spring Boot): Spring Boot, Spring Cloud Gateway, Spring Data JPA + Hibernate, Nimbus JOSE + JWT, Spring Security Crypto, mssql-jdbc driver.
- Web frontend (React): React, React Router, react-hook-form, i18next, Leaflet / react-leaflet, Recharts, GSAP, react-dropzone.
- Mobile app (Expo + React Native): expo-router, Zustand, expo-secure-store, TypeScript.
- Database and migrations: Microsoft SQL Server, Liquibase.
- API mock: json-server, jsonwebtoken, bcryptjs (mock-server.cjs).

5.2 System configuration

Table 5

Configuration files and variables per component

Component	Configuration file	Key variables
Microservices (iam, fleet-maintenance, booking-reservation)	application.properties + .env (from .env.example)	See Annex B. Environment variables (DATABASE_*, APP_*_PORT, JWT, etc.)
API Gateway	application.properties + .env (from .env.example)	GATEWAY_PORT, IAM_URL, FLEET_URL, BOOKING_URL, CORS_ALLOWED_ORIGINS, RATE_LIMIT_*
*-db repositories	liquibase.properties (from liquibase.properties.example)	url, username, password, parameter.<service>AppPassword

Component	Configuration file	Key variables
Web frontend	vite.config.js, .env	VITE_API_URL, VITE_MOCK_API_URL; proxy /api → gateway (port 8080) and /mock → mock-server (port 3100)
Mobile app	config/env.ts	EXPO_PUBLIC_API_URL
Containers	docker-compose.yml per repository	Shared external network rentamovil

Configuration files containing credentials (.env, liquibase.properties) are never pushed to the repository: each one is created locally from its matching .example file. The full detail of environment variables per component is in Annex B. Environment variables.

6. System architecture

6.1 Architectural style

The backend follows a microservices architecture organized by business domains (bounded contexts): identity and access, fleet and maintenance, reservations and rentals, payments and billing, and GPS telemetry. Each microservice is internally implemented with hexagonal architecture (ports and adapters), so that business rules do not depend on Spring, the database or HTTP.

The frontends use a feature-based organization with reusable components (component-driven): each module groups its pages, components, hooks and services.

All services share a single SQL Server database, but each one owns its own schema and connects with a user that only has permissions on that schema (ADR-009).

6.2 Architecture diagram

Figure 1

6.2 Architecture diagram

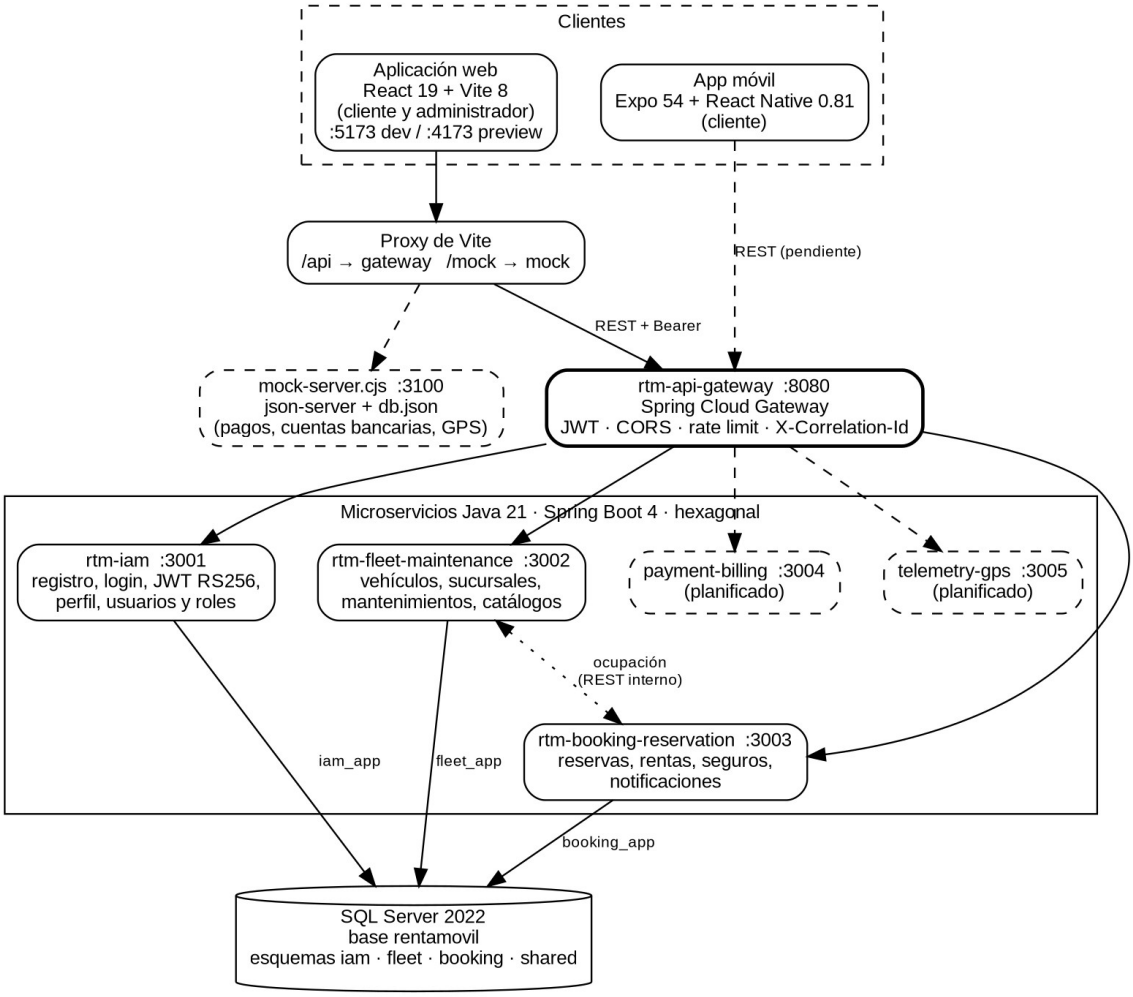


Figure 1. Container architecture. Dashed lines indicate planned services or pending integrations.

6.3 Components and ports

Table 6

System ports in local development

Component	Port	Note
Web frontend (Vite dev)	5173	npm run dev
Web frontend (preview / Docker)	4173	npm run preview or rentamovil-web container
rtm-api-gateway	8080	Single entry point for clients
rtm-iam	3001	Internal; only through the gateway
rtm-fleet-maintenance	3002	Internal
rtm-booking-reservation	3003	Internal
payment-billing (planned)	3004	—
telemetry-gps (planned)	3005	—
mock-server.cjs	3100	Payments, bank accounts and GPS while their services do not exist
SQL Server	1433	rentamovil-sqlserver container

6.4 Layers of each microservice (hexagonal)

The three implemented microservices share the same organization. The rule that cannot be broken: domain imports nothing from infrastructure nor from Spring; dependencies always point toward the domain.

Table 7

Layers of a microservice

Package	Responsibility
domain/entity	Business entities and value objects (Reservation, Vehicle, User...). They do not depend on Spring or JPA.
domain/enums	Domain statuses (ReservationStatus, VehicleStatus, UserStatus, MaintenanceStatus...).
domain/exception	Business exceptions (BusinessRuleException, ResourceNotFoundException...).
domain/port/in	Input ports: interfaces of the use cases the service offers.

Package	Responsibility
domain/port/out	Output ports: what the domain needs from the outside (repositories, token verifier, events, audit).
application/usecase	Use case implementation; orchestrates domain and ports.
infrastructure/web	Input adapters: REST controllers, DTOs, global error handler and filters.
infrastructure/persistence	Output adapters: JPA entities, Spring Data repositories and adapters implementing the *RepositoryPort.
infrastructure/security	JWT verification or issuance and password hashing.
infrastructure/config	BeanConfig: wires each use case with its adapters (dependency injection).

6.5 Communication between components

- **Client → backend: REST over HTTP with JSON. The browser calls /api (gateway) and /mock (mock) on its own origin; the Vite proxy forwards each to its destination.**
- **Gateway → services: the gateway routes by path prefix (section 9.1) and adds X-User-Id, X-User-Role and X-Correlation-Id.**
- **Service → service: internal REST calls; for example, fleet-maintenance queries booking-reservation for vehicle occupancy in order to calculate availability.**
- **Domain events: booking-reservation dispatches them in memory (InProcessEventDispatcher) and fleet publishes them in the log; between services they are delivered via POST /internal/events. No message broker is used at this scale.**

Table 8

Domain events

Event	Published by	Consumed by / effect
ReservationCreated / ReservationReturnBranchChanged / ReservationExpired	booking-reservation	In-app notifications (booking)
RentalStarted	booking-reservation	fleet-maintenance: vehicle to RENTED; telemetry-gps (future)
RentalCompleted	booking-reservation	fleet-maintenance: vehicle to AVAILABLE
PaymentCompleted / PaymentRejected	payment-billing (future)	booking-reservation: reservation to CONFIRMED or rejection; notification
MaintenanceCompleted	fleet-maintenance	booking-reservation: MAINTENANCE_COMPLETED notification

6.6 Path of a request

Example: the customer checks their reservations from the web.

1. The My Reservations screen calls reservationService.getMine(), which uses httpClient.get('/reservations/mine').
2. httpClient decides the base: /reservations is in REAL_BACKEND_PREFIXES, so it calls /api/reservations/mine with the access token in the Authorization: Bearer header.
3. The Vite proxy strips /api and forwards to http://localhost:8080/reservations/mine (without an Origin header).
4. The gateway assigns X-Correlation-Id, applies rate limiting, verifies the JWT signature and expiry with iam's public key, strips any external X-User-* headers and adds X-User-Id and X-User-Role.
5. The /reservations/** route is forwarded to booking-reservation (:3003).
6. In booking, the ReservationController re-verifies the token (CurrentUserArgumentResolver), calls the use case, and the use case calls the ReservationRepositoryPort.

7. The JPA adapter queries booking.reservation with the booking_app user and returns the rows.
8. The response returns through the same path; if the access token expired (401), httpClient requests a new token from /auth/refresh once and repeats the request.

7. Project structure

7.1 Microservice (example: rtm-booking-reservation)

```
rtm-booking-reservation/
├── src/main/java/com/rentamovil/booking/
│   ├── domain/ entity · enums · exception · port/in · port/out
│   ├── application/usecase/ use case implementations
│   └── infrastructure/
│   ├── config/ BeanConfig, WebConfig
│   ├── events/ InProcessEventDispatcher
│   ├── fleet/ availability adapter (provisional)
│   ├── persistence/ entity (JPA) · repository · adapter
│   ├── scheduling/ ReservationExpirationJob (expires unpaid reservations)
│   ├── security/ JwtKeyConfig, JwtVerifierAdapter
│   └── web/ controller · dto · exception · filter · auth
├── src/main/resources/application.properties
├── src/test/java/ unit and integration tests
├── .env.example · Dockerfile · docker-compose.yml · pom.xml · README.md
```

7.2 Database repository (example: rtm-iam-db)

```
rtm-iam-db/
├── 01_ddl/ 01_schemas · 03_tables · 04_alter · 05_views
│   10_indexes
├── 02_dml/ 00_inserts · 01_updates · 02_deletes
├── 03_dcl/ 00_roles · 01_grants
├── 04_tcl/
├── 05_rollbacks/
├── changelog/ master Liquibase changelog
├── liquibase.properties.example · docker-compose.yml
```

```
└── Dockerfile · pom.xml
```

Table 9

Folders of a database repository

Folder	Purpose
01_ddl	Schemas, tables, foreign keys (04_alter), views and indexes (10_indexes).
02_dml	Seed data (roles, permissions, categories, brands, models, maintenance types).
03_dcl	Service login and database user, and their permissions (GRANT).
04_tcl	Transaction control, when applicable.
05_rollbacks	Mirror with the rollback of each change.
changelog	Master changelog that Liquibase executes in order.
liquibase.properties	Connection and parameters (*_app user password). Copied from liquibase.properties.example and never uploaded.

7.3 Web frontend (RentaMovil/web/Front-end)

```
web/Front-end/
├── src/
│ ├── features/ auth · vehicles · booking · payment · notification
│ │ └── admin/ registerVehicle · VehicleInventory · registerMaintenance
│ │ historyMaintenance · reservations · branches
│ │ insuranceTypes · bankAccounts · users
│ │ vehicleLocation · routeHistory
│ ├── contexts/ AuthContext, ThemeContext
│ ├── shared/ api · components · hooks · services · utils
│ ├── Traductions/ es · en · fr · pt
│ ├── App.jsx protected routes with RequireAuth / RequireRole
│ └── main.jsx providers: Auth, Reservation, Payment
├── mock-server.cjs · db.json API mock (port 3100)
```

```
├── vite.config.js /api and /mock proxy
├── Dockerfile · docker-compose.yml · package.json
```

Table 10

Web frontend folders

Folder / file	Purpose
src/features/<module>/pages	Screens of each module (auth, vehicles, booking, payment, notification, admin/*).
src/features/<module>/ components	Components specific to the module.
src/features/<module>/hooks	Screen state and logic (useCars, useReservationsList, useCreateMaintenance...).
src/features/<module>/services	HTTP calls and mappers that convert the backend response into the view model.
src/contexts	Global contexts: AuthContext (session and user) and ThemeContext.
src/shared/api	httpClient.js (HTTP client with refresh) and tokenStore.js (access token in memory).
src/shared/components	Reusable components: layout (Navbar, Footer), ProtectedRoute (RequireAuth / RequireRole), dialogs.
src/shared/services and hooks	Cross-cutting services (cloudinaryService, useImageUpload, useIsMobile).
src/Traductions	i18next configuration and es, en, fr, pt files.
mock-server.cjs + db.json	API mock for the domains with no real service.
vite.config.js	/api → gateway (:8080) and /mock → mock (:3100) proxy.

7.4 Mobile app (RentaMovil/app_mobile/Frontend)

```
app_mobile/Frontend/src/
├── app/ expo-router routes: (tabs) · auth · account
│ reservation · payment · notifications
│ reservation-detail
```

```
├── features/ account · auth · branches · home · insurance
│   menu · notification · payment · reservation
│   vehicles (pages · components · services · mocks)
├── shared/ api (httpClient, tokenStore) · components · hooks
├── config/ env.ts (API URL per platform), assets.ts
├── theme/ themes and design constants
├── translation/ i18n es · en · fr · pt
└── types/ TypeScript domain types
```

8. Database

Aspect	Value
Engine	Microsoft SQL Server 2022 (rentamovil-sqlserver container, port 1433)
Database	rentamovil (shared by all services)
Schemas	iam (rtm-iam), fleet (rtm-fleet-maintenance), booking (rtm-booking-reservation), shared (common audit)
Application users	iam_app, fleet_app, booking_app: CRUD only on their schema + INSERT into shared.audit
Migrations	Liquibase 5.0.4, one repository per schema (rtm-iam-db, rtm-fleet-maintenance-db, rtm-booking-db)
Tables	21 tables: 7 in iam, 1 in shared, 9 in fleet and 4 in booking

8.1 Entity-relationship diagram

Figure 2

8.1 Entity-relationship diagram

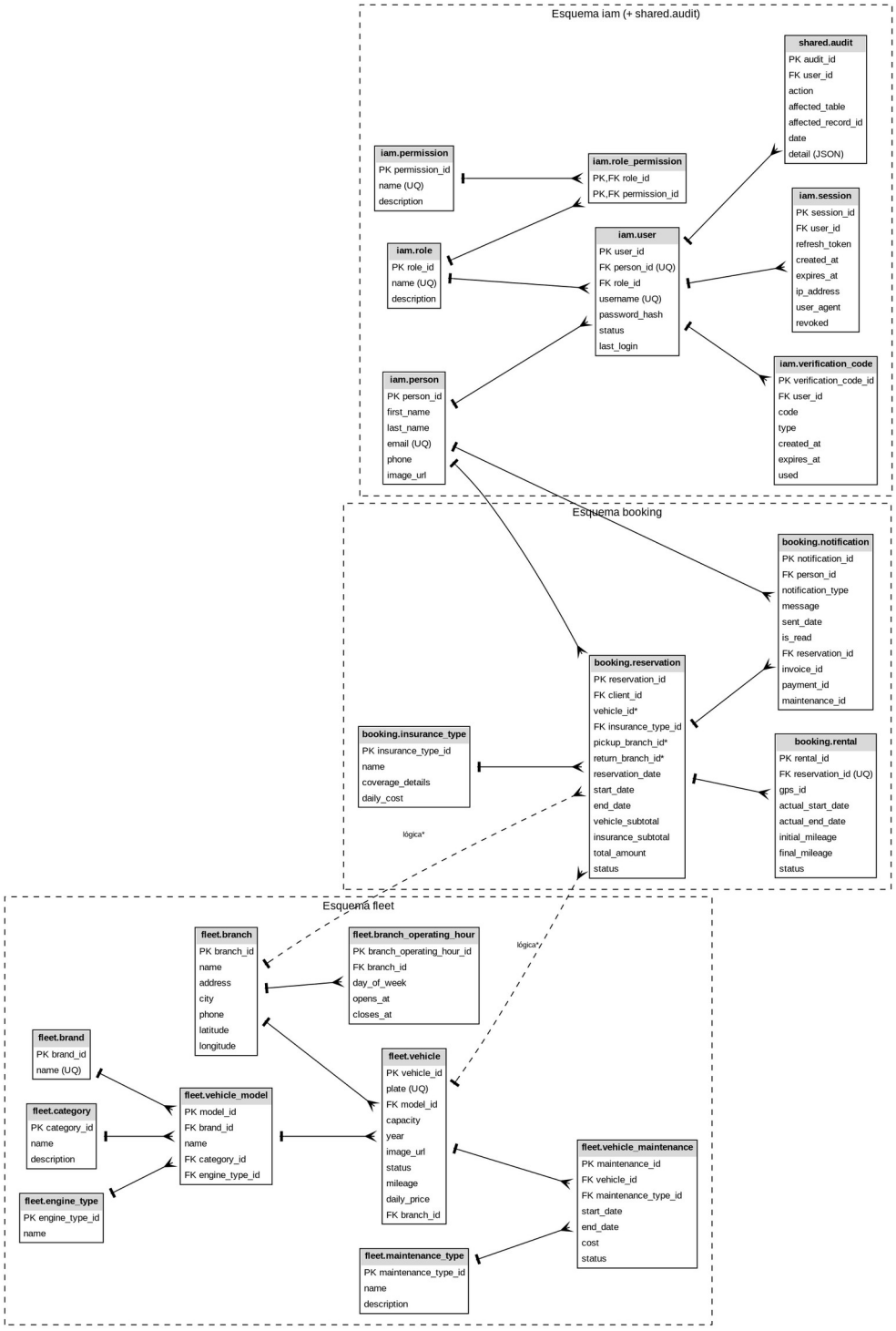


Figure 2. Entity-relationship model by schema. Dashed lines are logical references between schemas with no physical foreign key.

Between schemas there are only foreign keys to iam.person (reservation holder and notification recipient) and to iam.user (audit). The references from booking.reservation to fleet.vehicle and fleet.branch are logical: each service owns its data and validates them through the API, not through the database.

8.2 Main tables

Table	Purpose
iam.person / iam.user	Personal data and access account (1 to 1). The person is the one who reserves; the user is the one who signs in.
iam.role / iam.permission / iam.role_permission	CLIENT, ADMIN, SUPER_ADMIN roles and their 25 permissions.
iam.session / iam.verification_code	Sessions with refresh token and password recovery codes.
shared.audit	Log of administrative actions from all services.
fleet.vehicle and catalogs	Vehicles with model, brand, category, engine type, branch, status and daily rate.
fleet.branch / branch_operating_hour	Branches with coordinates and daily schedule.
fleet.vehicle_maintenance / maintenance_type	Maintenance and its type catalog.
booking.reservation	Reservation with dates, branches, insurance, subtotals, total and status.
booking.rental	Rental execution: pickup, return, mileages and GPS.
booking.insurance_type / notification	Insurance plans and in-app notifications.

8.3 Data dictionary

iam.person

Field	Type	Constraint	Description
person_id	INT	PK, IDENTITY	Person identifier.
first_name	VARCHAR(100)	NOT NULL	First names.
last_name	VARCHAR(100)	NOT NULL	Last names.
email	VARCHAR(150)	NOT NULL, UNIQUE	Access email; unique in the system.
phone	VARCHAR(20)	NULL	Contact phone.
image_url	VARCHAR(500)	NULL	Profile photo URL (Cloudinary).

iam.role

Field	Type	Constraint	Description
role_id	INT	PK, IDENTITY	Role identifier.
name	VARCHAR(50)	NOT NULL, UNIQUE	CLIENT, ADMIN or SUPER_ADMIN.
description	VARCHAR(255)	NULL	Role description.

iam.permission

Field	Type	Constraint	Description
permission_id	INT	PK, IDENTITY	Permission identifier.
name	VARCHAR(50)	NOT NULL, UNIQUE	Permission in resource:action format (e.g. vehicles:create).
description	VARCHAR(255)	NULL	Permission description.

iam.role_permission

Field	Type	Constraint	Description
role_id	INT	PK, FK → iam.role	Role.

Field	Type	Constraint	Description
permission_id	INT	PK, FK → iam.permission	Permission assigned to the role.

iam.user

Field	Type	Constraint	Description
user_id	INT	PK, IDENTITY	Account identifier (sub claim of the JWT).
person_id	INT	FK → iam.person, UNIQUE	Person owning the account (one account per person).
role_id	INT	FK → iam.role	Account role.
username	VARCHAR(50)	NOT NULL, UNIQUE	Username; cannot be modified.
password_hash	VARCHAR(255)	NOT NULL	BCrypt hash of the password.
status	VARCHAR(20)	CHECK ACTIVE/INACTIVE/BLOCKED	Account status.
last_login	DATETIME	NULL	Last successful sign-in.

iam.verification_code

Field	Type	Constraint	Description
verification_code_id	INT	PK, IDENTITY	Code identifier.
user_id	INT	FK → iam.user	Account that requested the code.
code	VARCHAR(10)	NOT NULL	6-digit code.
type	VARCHAR(30)	NOT NULL	Purpose (password recovery).

Field	Type	Constraint	Description
created_at	DATETIME	NOT NULL	Issued date.
expires_at	DATETIME	NOT NULL	Expiry (15 minutes by default).
used	BIT	NOT NULL	Indicates whether it was already used.

iam.session

Field	Type	Constraint	Description
session_id	INT	PK, IDENTITY	Session identifier.
user_id	INT	FK → iam.user, index	Session account.
refresh_token	VARCHAR(255)	NOT NULL	Active refresh token of the session.
created_at	DATETIME	NOT NULL	Session start.
expires_at	DATETIME	NOT NULL	Refresh token expiry (7 days by default).
ip_address	VARCHAR(45)	NULL	Source IP.
user_agent	VARCHAR(255)	NULL	Browser or client.
revoked	BIT	NOT NULL	Revoked session (logout, rotation or reuse detected).

shared.audit

Field	Type	Constraint	Description
audit_id	INT	PK, IDENTITY	Audit record identifier.
user_id	INT	FK → iam.user, NULL	Who performed the action (NULL = system or anonymous).
action	VARCHAR(50)	NOT NULL	Action (e.g. USER_UNBLOCKED, ROLE_GRANTED).

Field	Type	Constraint	Description
affected_table	VARCHAR(50)	NOT NULL, index	Affected table.
affected_record_id	INT	NULL, index	Affected record.
date	DATETIME	NOT NULL	Date and time.
detail	NVARCHAR(MAX)	CHECK ISJSON	Additional detail in JSON.

fleet.branch

Field	Type	Constraint	Description
branch_id	INT	PK, IDENTITY	Branch identifier.
name	VARCHAR(100)	NOT NULL	Location name.
address	VARCHAR(200)	NOT NULL	Physical address.
city	VARCHAR(100)	NOT NULL	City.
phone	VARCHAR(20)	NULL	Contact phone.
latitude	DECIMAL(10,8)	NULL	Latitude for the map.
longitude	DECIMAL(11,8)	NULL	Longitude for the map.

fleet.branch_operating_hour

Field	Type	Constraint	Description
branch_operating_hour_id	INT	PK, IDENTITY	Identifier.
branch_id	INT	FK → fleet.branch	Branch.
day_of_week	SMALLINT	CHECK 1..7; UNIQUE with branch_id	Day (1 = Monday … 7 = Sunday).
opens_at	TIME	NOT NULL	Opening time.
closes_at	TIME	NOT NULL	Closing time.

fleet.brand

Field	Type	Constraint	Description
brand_id	INT	PK, IDENTITY	Brand identifier.
name	VARCHAR(50)	NOT NULL, UNIQUE	Brand name.

fleet.category

Field	Type	Constraint	Description
category_id	INT	PK, IDENTITY	Category identifier.
name	VARCHAR(50)	NOT NULL	Name (Sedan, SUV...).
description	VARCHAR(255)	NULL	Description.

fleet.engine_type

Field	Type	Constraint	Description
engine_type_id	INT	PK, IDENTITY	Identifier.
name	VARCHAR(50)	NOT NULL	Engine or fuel type.

fleet.vehicle_model

Field	Type	Constraint	Description
model_id	INT	PK, IDENTITY	Model identifier.
brand_id	INT	FK → fleet.brand	Brand.
name	VARCHAR(50)	NOT NULL	Model name.
category_id	INT	FK → fleet.category	Category.
engine_type_id	INT	FK → fleet.engine_type	Engine type.

fleet.vehicle

Field	Type	Constraint	Description
vehicle_id	INT	PK, IDENTITY	Vehicle identifier.
plate	VARCHAR(10)	NOT NULL, UNIQUE	License plate.
model_id	INT	FK → fleet.vehicle_model	Model (brand, category and engine are derived from it).
capacity	INT	NULL	Number of passengers.
year	INT	NULL	Model year.
image_url	VARCHAR(255)	NULL	Photo URL.
status	VARCHAR(30)	CHECK AVAILABLE/RENTED/MAINTENANCE/RETIRED; index	Operational status.
mileage	DECIMAL(10,2)	NOT NULL	Current mileage.
daily_price	DECIMAL(10,2)	NOT NULL	Daily rate (COP).
branch_id	INT	FK → fleet.branch; index	Branch it belongs to.

fleet.maintenance_type

Field	Type	Constraint	Description
maintenance_type_id	INT	PK, IDENTITY	Identifier.

Field	Type	Constraint	Description
name	VARCHAR(50)	NOT NULL	Type (Oil change, Brake inspection...).
description	VARCHAR(255)	NULL	Description.

fleet.vehicle_maintenance

Field	Type	Constraint	Description
maintenance_id	INT	PK, IDENTITY	Maintenance identifier.
vehicle_id	INT	FK → fleet.vehicle; index with status	Serviced vehicle.
maintenance_type_id	INT	FK → fleet.maintenance_type	Maintenance type.
start_date	DATE	NOT NULL	Start date.
end_date	DATE	NULL	End date (NULL while it is open).
cost	DECIMAL(10, 2)	NULL	Cost.
status	VARCHAR(30)	CHECK SCHEDULED/IN_PROGRESS/COMPLETED	Maintenance status.

booking.insurance_type

Field	Type	Constraint	Description
insurance_type_id	INT	PK, IDENTITY	Insurance plan identifier.
name	VARCHAR(100)	NOT NULL	Coverage name.

Field	Type	Constraint	Description
coverage_details	TEXT	NULL	What it includes and excludes.
daily_cost	DECIMAL(10,2)	NOT NULL, CHECK >= 0	Cost per day.

booking.reservation

Field	Type	Constraint	Description
reservation_id	INT	PK, IDENTITY	Reservation identifier.
client_id	INT	FK → iam.person; index	Reservation holder (personId claim).
vehicle_id	INT	NOT NULL (logical reference a fleet.vehicle)	Reserved vehicle.
insurance_type_id	INT	FK → booking.insurance_type, NULL	Chosen insurance (optional).
pickup_branch_id	INT	NOT NULL (logical reference a fleet.branch)	Pickup branch.
return_branch_id	INT	NOT NULL (logical reference a fleet.branch)	Return branch.

Field	Type	Constraint	Description
reservation_date	DATETIME	NOT NULL	Creation date.
start_date	DATETIME	NOT NULL	Start of the rental.
end_date	DATETIME	NOT NULL	End of the rental.
vehicle_subtotal	DECIMAL(10,2)	CHECK >= 0	Vehicle rental value.
insurance_subtotal	DECIMAL(10,2)	CHECK >= 0	Insurance value.
total_amount	DECIMAL(10,2)	CHECK = vehicle_subtotal + insurance_subtotal	Reservation total.
status	VARCHAR(30)	CHECK PENDING_PAYMENT/PENDING_REVIEW/CONFIRMED/CANCELLED/COMPLETED	Reservation status.

booking.rental

Field	Type	Constraint	Description
rental_id	INT	PK, IDENTITY	Rental identifier.
reservation_id	INT	FK → booking.reservation, UNIQUE	Reservation being executed (one rental per reservation).
gps_id	INT	NOT NULL	GPS device assigned at pickup.
actual_start_date	DATETIME	NOT NULL	Actual pickup date.

Field	Type	Constraint	Description
actual_end_date	DATETIME	NULL	Actual return date.
initial_mileage	DECIMAL(10,2)	CHECK >= 0	Starting mileage.
final_mileage	DECIMAL(10,2)	NULL	Ending mileage.
status	VARCHAR(30)	CHECK IN_PROGRESS/COMPLETED; COMPLETED requires final date and mileage	Rental status.

booking.notification

Field	Type	Constraint	Description
notification_id	INT	PK, IDENTITY	Identifier.
person_id	INT	FK → iam.person; index with is_read	Recipient.
notification_type	VARCHAR(50)	CHECK (8 types)	RESERVATION_CREATED, RESERVATION_CANCELLED, RESERVATION_RETURN_BRANCH_CHANGED, RESERVATION_RETURN_DATE_EXTENDED, RESERVATION_EXPIRED, PAYMENT_APPROVED, PAYMENT_REJECTED or MAINTENANCE_COMPLETED.
message	TEXT	NOT NULL	Notice text.
sent_date	DATETIME	NOT NULL	Sent date.
is_read	BIT	NOT NULL	Read by the recipient.
reservation_id	INT	FK → booking.reservation, NULL	Related reservation.

Field	Type	Constraint	Description
invoice_id	INT	NULL	Related invoice (payment-billing).
payment_id	INT	NULL	Related payment (payment-billing).
maintenance_id	INT	NULL	Related maintenance (fleet).

8.4 Notable integrity rules

- booking.reservation: total_amount = vehicle_subtotal + insurance_subtotal, and both subtotals >= 0 (CHECK).
- booking.rental: one rental per reservation (UNIQUE); a COMPLETED rental requires final date and mileage.
- fleet.vehicle: unique plate; status restricted to AVAILABLE, RENTED, MAINTENANCE or RETIRED.
- fleet.branch_operating_hour: a single schedule per branch and day (UNIQUE branch_id + day_of_week).
- iam.user: one user per person, unique username; status ACTIVE, INACTIVE or BLOCKED. iam.person: unique email.
- shared.audit.detail must be valid JSON (CHECK ISJSON).

9. Technical modules

rtm-api-gateway

Element	Detail
Purpose	Single entry point: prefix routing, JWT verification, CORS, rate limiting, X-Correlation-Id, handling of downed services (503/504) and an aggregated GET /health.
Related files	application.properties (routes), filter/JwtAuthenticationFilter, filter/RateLimitFilter, filter/CorrelationIdFilter, filter/UpstreamErrorFilter, security/JwtVerifier, ratelimit/InMemoryRateLimiter, health/HealthController, config/CorsConfig
Related tables	None (it has no database)
Main operations	Forward requests; validate tokens; limit requests; report service status.

rtm-iam (Identity and access)

Element	Detail
Purpose	Registration, login, RS256 JWT issuance, refresh with rotation, logout, password recovery, profile, email and password change, user and role administration.
Related files	web/controller: AuthController, PasswordController, ProfileController, UserAdminController, JwksController, HealthController; application/usecase: LoginUseCaseImpl, RegisterUserUseCaseImpl, RefreshTokenUseCaseImpl, ResetPasswordUseCaseImpl, ChangeUserStatusUseCaseImpl...; infrastructure/security: JwtTokenProviderAdapter, BCrypt; infrastructure/email: LogEmailSenderAdapter
Related tables	iam.person, iam.user, iam.role, iam.permission, iam.role_permission, iam.session, iam.verification_code, shared.audit
Main operations	Register; authenticate and lock after 5 failures; renew and revoke sessions; recover password; update profile; change user role and status.

rtm-fleet-maintenance (Fleet)

Element	Detail
Purpose	Single source of truth for vehicles, their status and availability, the branches with schedule and coordinates, the maintenance records and the reference catalogs.
Related files	web/controller: VehicleController, BranchController, MaintenanceController, ReferenceDataController, InternalEventController; usecase: RegisterVehicleUseCaseImpl, SearchVehiclesUseCaseImpl, CheckVehicleAvailabilityUseCaseImpl, ChangeVehicleStatusUseCaseImpl, CompleteMaintenanceUseCaseImpl, ManageBranchUseCaseImpl, HandleRentalEventUseCaseImpl; client/BookingOccupancyClientAdapter
Related tables	fleet.vehicle, fleet.vehicle_model, fleet.brand, fleet.category, fleet.engine_type, fleet.branch, fleet.branch_operating_hour, fleet.vehicle_maintenance, fleet.maintenance_type, shared.audit
Main operations	Search and query vehicles; calculate availability; register and edit vehicles; send to maintenance and complete it; manage branches; react to RentalStarted/RentalCompleted.

rtm-booking-reservation (Reservations, rentals and notifications)

Element	Detail
Purpose	Reservation lifecycle (creation, query, return branch change, cancellation and automatic expiration after 24 h without payment), vehicle pickup and return, insurance plans and in-app notifications.
Related files	web/controller: ReservationController, RentalController, AdminBookingController, InsuranceController, NotificationController, InternalEventController, InternalReservationController; scheduling/ReservationExpirationJob; events/InProcessEventDispatcher; fleet/PendingVehicleAvailabilityAdapter
Related tables	booking.reservation, booking.rental, booking.insurance_type, booking.notification, shared.audit

Element	Detail
Main operations	Create, query, modify and cancel reservations; expire reservations; register pickup and return; list and create insurance plans; notify the customer; receive payment events.

Web frontend

Element	Detail
Purpose	Customer and administrator interface.
Related files	src/App.jsx (routes), contexts/AuthContext.jsx, shared/api/httpClient.js, shared/components/ProtectedRoute.jsx, features/*/services/*Service.js and *Mapper.js
Related tables	Consumes the APIs; does not access the database
Main operations	Authentication with automatic refresh; role-based routing; consumption of the gateway and the mock; image upload to Cloudinary; internationalization.

Mobile app

Element	Detail
Purpose	Mobile client (Customer role) with Expo Router.
Related files	src/app/* (routes), src/config/env.ts, src/shared/api/httpClient.ts, src/features/*/services/*.ts
Related tables	Consumes the APIs; does not access the database
Main operations	Login and registration, search, reservation, payment with receipt (camera or gallery), history, notifications and account. Pending connection to the gateway.

mock-server.cjs

Element	Detail
Purpose	Simulated API with json-server for the domains whose service does not exist.

Element	Detail
Related files	web/Front-end/mock-server.cjs, web/Front-end/db.json
Related tables	db.json collections: payments, bankAccounts, gps, locations (and old test data)
Main operations	Serve payments, bank accounts and GPS positions until payment-billing and telemetry-gps exist.

10. System API or routes

The full contracts (parameters, bodies and responses) are in OpenAPI 3 in rtm-docs/07-api/contracts/openapi/. The following tables list the implemented endpoints, as they are called through the gateway.

10.1 Gateway routing

The client calls	Is forwarded to
/auth/**	rtm-iam, stripping /auth (e.g. /auth/login → /login)
/users/**	rtm-iam, unchanged
/vehicles/**, /branches/**	rtm-fleet-maintenance (:3002)
/reservations/**, /rentals/**, /notifications/**, /insurance-plans/**, /admin/reservations/**, /admin/rentals/**, /admin/insurance-plans/**	rtm-booking-reservation (:3003)
/payments/**, /invoices/**, /bank-accounts/**	payment-billing (:3004, planned)
/tracking/**	telemetry-gps (:3005, planned)

Note: the web frontend also calls /maintenances and the catalogs (/brands, /categories, /engine-types, /vehicle-models, /maintenance-types), which fleet-maintenance exposes but which the verified gateway routing table does not yet include; they must be added to the fleet-maintenance route.

10.2 rtm-iam (the /auth prefix is stripped at the gateway; /users is forwarded unchanged)

Method	Path	Description	Access / permission
POST	/auth/register	Registers a customer and signs them in.	Public
POST	/auth/login	Authenticates with email or username; returns access and refresh tokens.	Public
POST	/auth/refresh	Renews the access token and rotates the refresh token.	Public (refresh token)
POST	/auth/logout	Revokes the session.	Session
GET	/auth/jwks	RS256 public key for verifying tokens.	Public
POST	/auth/password/forgot	Generates and sends the recovery code.	Public
POST	/auth/password/reset	Changes the password with the code.	Public
GET	/users/me	Profile of the authenticated user.	Session
PATCH	/users/me	Updates first name, last name, phone or photo.	Session
PATCH	/users/me/email	Changes the email (requires the current password).	Session
PATCH	/users/me/password	Changes the password and revokes all sessions.	Session
GET	/users	Lists users.	users:read
PATCH	/users/{id}/role	Changes a user's role.	roles:grant
PATCH	/users/{id}/status	Unlocks (BLOCKED→ACTIVE) or deactivates (ACTIVE→INACTIVE).	users:update-status

10.3 rtm-fleet-maintenance

Method	Path	Description	Access / permission
GET	/vehicles	Paginated catalog search (dates, branch, brand, category, engine, price).	Public
GET	/vehicles/inventory	Full inventory (any status).	Administration
GET	/vehicles/stats	Fleet indicators.	Administration
GET	/vehicles/{id}	Detail of a vehicle.	Public
GET	/vehicles/{id}/availability?from&to	Availability over a date range.	Public
POST	/vehicles	Registers a vehicle.	vehicles:create
PUT	/vehicles/{id}	Edits a vehicle.	vehicles:update
PATCH	/vehicles/{id}/status	Changes the status; to MAINTENANCE it opens a maintenance record.	vehicles:update / maintenance:create
POST	/vehicles/{id}/maintenance/complete	Closes the ongoing maintenance.	maintenance:update
GET	/maintenances	Maintenance history.	Administration
GET	/branches	Lists branches.	Public
POST	/branches	Creates a branch.	branches:manage
PUT	/branches/{id}	Edits a branch.	branches:manage
DELETE	/branches/{id}	Deletes a branch with no vehicles.	branches:manage
GET	/brands, /categories, /engine-types, /vehicle-models, /maintenance-types	Reference catalogs.	Public / session

10.4 rtm-booking-reservation

Method	Path	Description	Access / permission
POST	/reservations	Creates a reservation (PENDING_PAYMENT).	reservations:create
GET	/reservations/mine	Reservations of the authenticated user.	reservations:read:own
GET	/reservations/{id}	Detail of a reservation.	Owner or reservations:read
PATCH	/reservations/{id}	Changes the return branch.	reservations:update(:own)
POST	/reservations/{id}/cancel	Cancels the reservation.	reservations:cancel(:own)
GET	/reservations/admin	All reservations with filters.	reservations:read
GET	/reservations/insurance-types	Insurance plans.	Public
POST	/reservations/insurance-types	Creates an insurance plan.	insurance-types:manage
POST	/rentals/{reservationId}/pickup	Records the pickup (GPS and mileage).	rentals:manage
POST	/rentals/{id}/return	Records the return and completes the reservation.	rentals:manage
GET	/rentals/in-progress, /rentals/admin	Ongoing rentals and all rentals.	Administration
GET	/notifications	User notifications.	Session
PATCH	/notifications/{id}/read	Marks a notification as read.	Session

10.5 Internal use between services (not exposed by the gateway)

Method	Path	Description	Access / permission
GET	booking: /internal/reservations/vehicle-occupancy	Vehicle occupancy; the fleet query to calculate availability.	Internal
POST	booking and fleet: /internal/events	Receives domain events (PaymentCompleted, RentalStarted...).	Internal
GET	/health (each service and the gateway)	Service and database status.	Public

10.6 mock-server.cjs (port 3100, /mock prefix in the frontend)

Method	Path	Description	Access / permission
GET/PATCH	/payments, /payments/{id}	Payments and their approval or rejection (replaces payment-billing).	—
GET/POST/PATCH	/bank-accounts	Collection bank accounts.	—
GET	/gps, /locations	GPS devices and positions (replaces telemetry-gps).	—

10.7 Response codes and error format

All services respond to errors with the same shape:

```
{ "error": "VALIDATION_ERROR",
"message": "The email field is required",
"details": [ { "field": "email", "message": "The email field is required" } ] }
```

The error field is stable and is the one the frontend uses to decide; message is in Spanish for the user. Paginated lists use page (1 by default) and limit (20 by default, max 100) and respond { items, page, limit, total, totalPages }; an empty list is not an error.

Code	Cause	What to do
200 / 201 / 204	Success; 201 returns the created resource; 204 has no body.	—
400 VALIDATION_ERROR	Data with invalid shape or a rule expressed on a field.	Fix the request; details indicates the field.
401 INVALID_TOKEN	No token, altered token or expired token.	The frontend attempts a refresh; if it fails, it returns to login.
403	Valid token without the required permission; or CORS not allowed.	Check role/permissions or CORS_ALLOWED_ORIGINS.
404	The resource does not exist.	Verify the identifier.
409	Conflict with the current state (e.g. FLEET_SERVICE_UNAVAILABLE, vehicle unavailable, branch with vehicles).	Check the resource status or the dependent service.
423	Account locked after 5 failed attempts.	A SUPER_ADMIN unlocks it.
429 RATE_LIMITED	The gateway limit was exceeded; it includes Retry-After.	Wait the indicated seconds.
503 SERVICE_UNAVAILABLE	The target service is not running.	Bring up the service; check the gateway GET /health.
504 UPSTREAM_TIMEOUT	The target service did not respond in time.	Check the service and database logs.

11. Security

Aspect	Implementation
Authentication	JWT signed with RS256 by rtm-iam. Claims: sub (user_id), email, role, permissions and personId. The access token lasts 30 min (APP_IAM_JWT_EXPIRY=1800) and the refresh token 7 days (604800 s).
Key publication	rtm-iam publishes its public key at /jwks. The gateway downloads and caches it; if it changes, it requests it again. Each service re-verifies the token with the public key before authorizing.
Sessions and refresh	Each login creates a row in iam.session. The refresh rotates the token: the previous one is revoked. If an already-used refresh arrives (possible theft), iam revokes all of the user's sessions. Changing or resetting the password also revokes all sessions.
Passwords	BCrypt hash with 12 rounds (APP_IAM_BCRYPT_ROUNDS). Policy in the web frontend: minimum 8 characters, uppercase, number and special character.
Account locking	On the 5th consecutive failed attempt the account moves to BLOCKED and login responds 423. Only a SUPER_ADMIN unlocks it (PATCH /users/{id}/status).
Password recovery	6-digit code that expires in 15 minutes (APP_IAM_PASSWORD_RESET_CODE_EXPIRY=900). The endpoint responds the same way whether or not the email exists, in order not to reveal accounts.
Authorization (RBAC)	Three roles (CLIENT, ADMIN, SUPER_ADMIN) and 25 resource:action permissions in iam.role_permission. Permissions travel in the token; the frontend hides screens with RequireRole, but each service is what protects the data.
Gateway	It always strips incoming external X-User-* headers and sets them itself from the verified token. Rate limiting: 10 requests every 5 min per IP on login, registration and recovery; 100 per minute elsewhere. CORS only for CORS_ALLOWED_ORIGINS.
Traceability	The gateway assigns or respects X-Correlation-Id and propagates it to the services and to the response, so a request can be followed across all logs.

Aspect	Implementation
Database	Each service connects with its own user (iam_app, fleet_app, booking_app) that only has SELECT/INSERT/UPDATE/DELETE on its schema and INSERT on shared.audit; never with sa. Hibernate runs with ddl-auto=validate: the service does not create or alter tables.
Audit	Administrative actions (unlocking, deactivation, role change, fleet and reservation operations) are recorded in shared.audit with JSON detail.
Secrets	Passwords, JWT keys and connection strings are read from .env / liquibase.properties, which are never uploaded (there are .example files). Each environment must have its own RS256 key pair.
Frontend	The access token lives only in memory (tokenStore); the refresh token is stored in localStorage. When the refresh fails, the session is cleared and it returns to login.
Files	Images are uploaded to Cloudinary; iam only accepts profile URLs with the configured prefix (APP_IAM_PROFILE_IMAGE_ALLOWED_PREFIX).

11.1 Permissions by role

Role	Permissions
CLIENT	vehicles:read, reservations:create, reservations:read:own, reservations:update:own, reservations:cancel:own, payments:create:own, invoices:read:own
ADMIN	The customer's ones + vehicles:create, vehicles:update, branches:manage, maintenance:create, maintenance:update, reservations:read, reservations:update, reservations:cancel, insurance-types:manage, rentals:manage, payments:read, payments:review, invoices:read, tracking:read, users:read
SUPER_ADMIN	All permissions: the administrator's ones + bank-accounts:manage, users:update-status and roles:grant

Source: rtm-iam-db/02_dml/00_inserts (002_insert_permissions.sql and 003_insert_role_permissions.sql).

12. Version control

All repositories use Git and are on GitHub (RentaMovil organization; rtm-docs under code-sena). Each repository follows the same convention (rtm-docs/00-governance/git-conventions.md):

- **Permanent branches: develop (integration) → qa (functional testing) → staging (production simulation) → main (production). Changes are promoted in that order.**
- **Working branches: feat/<description>, fix/<description>, chore/<description>, created from develop and merged via pull request; they are deleted after the merge.**
- **Hotfix: hotfix/<description> from main; merged into main and then towards staging, qa and develop.**
- **Commits: Conventional Commits (feat, fix, docs, refactor, test, chore), for example feat: add personId claim to the access token.**
- **One branch, one task. Different features are not mixed on the same branch.**

12.1 Architecture decisions (ADR)

ADR	Decision
ADR-001	Documentation language (technical in English, product in Spanish).
ADR-002	Decomposition into microservices.
ADR-003	Mobile app architecture.
ADR-004	State management.
ADR-005	Web authentication and authorization.
ADR-006	Web frontend architecture.
ADR-007	Frontend–backend boundary.
ADR-008	MVP route boundary.

ADR	Decision
ADR-009	Schema per service and migration repositories.

13. Maintenance recommendations

Task	Recommended procedure
Add a use case	1) Declare the interface in domain/port/in. 2) If it needs something external, declare the port in domain/port/out. 3) Implement it in application/usecase without importing Spring. 4) Implement the adapter in infrastructure. 5) Register the bean in BeanConfig. 6) Expose it in an infrastructure/web controller. 7) Write its tests and run mvn test.
Change the schema	First a new changeset in the *-db repository (never edit one already applied), with its rollback script in 05_rollbacks; apply it with mvn liquibase:update. Then adjust the JPA entity: since ddl-auto=validate, the service will not start if they do not match.
Add an endpoint	Define it first in the OpenAPI contract in rtm-docs/07-api/contracts (API first), respecting plural names, kebab-case, page/limit pagination and the common error format. If it is a new route for the client, add it to the gateway routing table (application.properties).
Connect a mock domain to the real backend	When the service exists, add its prefix to REAL_BACKEND_PREFIXES in web/Front-end/src/shared/api/httpClient.js; the rest of the frontend does not change.
Dependencies	Update versions in pom.xml / package.json on a chore/ branch, run tests and lint (mvn test, npm run lint) and validate on develop before promoting.
Database backup	BACKUP DATABASE rentamovil TO DISK = '/var/opt/mssql/backup/rentamovil_AAAA-MM-DD.bak' from sqlcmd in the rentamovil-sqlserver container; copy the file out of the container and store it off the server. Schedule at least one daily backup and one before each migration.
Restoration	Stop the services, run RESTORE DATABASE rentamovil FROM DISK = '...' WITH REPLACE, verify with each service's GET /health and check that the Liquibase migrations are up to date (mvn liquibase:status).
Observability	Follow a request through its X-Correlation-Id in the gateway and service logs; check the gateway GET /health, which reports the status of each service (ok / degraded / down).

13.1 Troubleshooting

Symptom	Likely cause	Solution
The gateway responds 503 SERVICE_UNAVAILABLE	The target service is not running, or the URL in .env is incorrect.	Bring up the service and check IAM_URL, FLEET_URL, BOOKING_URL; consult the gateway GET /health.
All authenticated requests respond 401 INVALID_TOKEN	The gateway or the service does not have the correct public key, or the machine clock is out of sync.	Verify that iam is up (/auth/jwks) and that APP_*_JWT_PUBLIC_KEY matches iam's pair.
A service does not start: schema validation error	The JPA entity does not match the table (ddl-auto=validate).	Apply the pending migrations from the *-db repository or fix the entity.
Creating a reservation responds 409 FLEET_SERVICE_UNAVAILABLE	booking-reservation still uses its provisional fleet adapter.	Expected behaviour until booking is connected to fleet-maintenance.
The browser shows a CORS error	The frontend origin is not in CORS_ALLOWED_ORIGINS.	Add the origin (e.g. http://localhost:5173) to the gateway .env.
Login responds 429	Gateway rate limiting (10 attempts every 5 min per IP).	Wait the Retry-After time; in development RATE_LIMIT_* can be adjusted.
Liquibase fails with 'Missing parameter.*AppPassword'	The service user password is missing in liquibase.properties.	Copy liquibase.properties.example and fill in the parameter.
The recovery code does not reach the email	In development iam uses LogEmailSenderAdapter: the code is written to the service log.	Search '[EMAIL DE DESARROLLO]' in the rtm-iam log; in production a real email adapter must be implemented.
Payments, bank accounts or GPS with no data on the web	The mock (port 3100) is not running.	Run npm run mock-api in web/Front-end or bring up the rentamovil-mock-api container.

13.2 Technical debt and known limitations

- booking-reservation uses a provisional fleet adapter: creating reservations responds 409 until it is connected to fleet-maintenance.
- payment-billing and telemetry-gps do not exist; the frontend uses the mock for payments, bank accounts and GPS. The action after rejecting a payment (re-upload or cancel) is not sent yet.
- The mobile app still points to the old mock (port 3001 and /auth/* routes that today belong to iam) and uses local data for branches, insurance, reservations and payments.
- The gateway rate limiting keeps counters in memory; with several gateways a RateLimiter over Redis must be implemented.
- iam email sending is a development adapter that writes the code to the log; in production it must be replaced by an email provider.
- The gateway routing table does not include /maintenances nor the reference catalogs consumed by the web frontend.
- Images are uploaded from the browser to Cloudinary with an unsigned preset; it is better to move it to environment variables or sign it from the backend.

14. Glossary

Term	Definition
API Gateway	Service that receives all client requests and forwards them to the corresponding microservice.
Microservice	Independent application that implements a business domain and is deployed separately.
Hexagonal architecture	Style that isolates the domain from the outside through ports (interfaces) and adapters (implementations).
Port / adapter	Interface that defines what the domain offers or needs / class that implements it with a specific technology.
JWT	JSON Web Token: signed credential that identifies the user and their permissions.
RS256	JWT signing with an RSA key pair: iam signs with the private key and the others verify with the public one.
JWKS	JSON Web Key Set: endpoint that publishes the public key for verifying tokens.
Refresh token	Long-lived credential that allows obtaining a new access token without signing in again.
RBAC	Access control based on roles and permissions.
Rate limiting	Limit of requests per period to protect the system from abuse.
CORS	Browser mechanism that restricts which origins may call an API.
Liquibase / changeset	Migration tool / each versioned schema change.
Schema	Logical grouping of tables inside the rentamovil database (iam, fleet, booking, shared).
DDL / DML / DCL	Structure, data and permission definition statements respectively.
Domain event	Message announcing that something happened in a service (e.g. RentalStarted) so that others react.
Mock	Simulated server that temporarily replaces a service that does not exist yet.
X-Correlation-Id	Identifier that accompanies a request across all services so it can be traced in the logs.

Term	Definition
OpenAPI	Standard specification for describing REST API contracts.

15. References

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16. Annexes

Annex A. Quick reference commands

Task	Command
Create the shared network and SQL Server + iam migrations	cd rtm-iam-db && docker compose up -d
Apply a schema's migrations	cd rtm-<service>-db && mvn liquibase:update
View pending migrations	mvn liquibase:status
Run a service's tests	mvn test
Bring up a service locally	mvn spring-boot:run
Bring up a service with Docker	docker compose up -d --build
Verify a service	curl http://localhost:3001/health (3002, 3003, 8080 depending on the service)
Web frontend in development	cd web/Front-end && npm install && npm run dev
API mock	npm run mock-api
Lint the web frontend	npm run lint
Mobile app	cd app_mobile/Frontend && npm install && npx expo start
Type-check the mobile app	npm run typecheck

Annex B. Environment variables

Component	Variables
Gateway	GATEWAY_PORT, IAM_URL, FLEET_URL, BOOKING_URL, PAYMENT_URL, TELEMETRY_URL, CORS_ALLOWED_ORIGINS, RATE_LIMIT_SENSITIVE_LIMIT/WINDOW, RATE_LIMIT_GENERAL_LIMIT/WINDOW

Component	Variables
rtm-iam	DATABASE_URL, DATABASE_USERNAME (iam_app), DATABASE_PASSWORD, APP_IAM_PORT, APP_IAM_JWT_PRIVATE_KEY, APP_IAM_JWT_PUBLIC_KEY, APP_IAM_JWT_EXPIRY, APP_IAM_REFRESH_TOKEN_EXPIRY, APP_IAM_BCRYPT_ROUNDS, APP_IAM_PASSWORD_RESET_CODE_EXPIRY, APP_IAM_PROFILE_IMAGE_ALLOWED_PREFIX
rtm-fleet-maintenance	DATABASE_* (fleet_app), APP_FLEET_MAINTENANCE_PORT, IAM_URL, BOOKING_URL, BOOKING_TIMEOUT, RETIRE_RESERVATION_HORIZON_DAYS
rtm-booking-reservation	DATABASE_* (booking_app), APP_BOOKING_RESERVATION_PORT, APP_BOOKING_RESERVATION_JWT_PUBLIC_KEY, APP_BOOKING_RESERVATION_PAYMENT_WINDOW_HOURS (24)
*-db repositories	MSSQL_SA_PASSWORD, MSSQL_PORT, MSSQL_DATABASE (rentamovil); in liquibase.properties: url, username, password and parameter.<service>AppPassword
Web frontend	VITE_API_URL (/api), VITE_MOCK_API_URL (/mock); in Docker: GATEWAY_URL, MOCK_URL
Mobile app	EXPO_PUBLIC_API_URL

Real values are never published: each .env and liquibase.properties is created from its .example and is not uploaded to the repository.

Annex C. Related documents

Document	Contents
User Manual	Use of the system by the customer (web and mobile).
Administrator Manual	Use of the administration panel.
Installation Manual	Requirements, installation, configuration and verification step by step.
rtm-docs/05-architecture	Architecture, ADR and client architecture.
rtm-docs/06-data	Models, dictionary and migration strategy.

Document	Contents
rtm-docs/07-api	API, authentication and OpenAPI contract guides.
rtm-docs/09-microservices	Service catalog and per-service documentation.
Each repository's README	Local usage, configuration and status of each service.

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